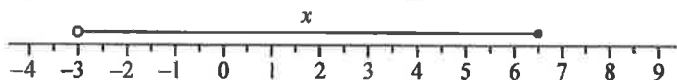


Paper 1: Basic

1. Solve $6x + 7 = -8$. [1]
[N21/I/9(a)]
2. (a) Expand and simplify $(2x + 3p)^2$. [2]
(b) Given that $(2x + 3p)^2 = 4x^2 - 36x + 81$, find the value of p . [1]
[N21/I/15]
3. Using factorisation, solve $8x^2 - 6x - 9 = 0$. [3]
[N21/I/21(b)]
4. Solve the equation $\frac{x}{5} - \frac{2x-3}{4} = -3$. [3]
[N20/I/9]
5. Use factorisation to solve the equation.
 $3p^2 + p - 10 = 0$ [3]
[N19/I/8]
6. Solve the inequalities $6 \leq 2x + 11 < 19$. [2]
[N19/I/9(b)]
7. A range of values for x is represented on the number line below.


Write down inequalities that represent this range of values for x . [1]
[N17/I/2]
8. Solve $\frac{3x-4}{2} - \frac{2x}{3} = 1$. [3]
[N17/I/11]
9. Solve $\frac{x}{5} + 14 = 8$. [1]
[N15/I/1]
10. (a) Solve the inequalities $-10 \leq 7 - 2x \leq 1$. [2]
(b) Write down all the integers that satisfy $-10 \leq 7 - 2x < 1$. [1]
[N15/I/15]

11. The mass of a small pot is a kg.

The mass of a large pot is b kg.

Ajay buys 4 small pots and 2 large pots with a total mass of 119 kg.

Bhanu buys 5 small pots and 3 large pots with a total mass of 165 kg.

Form and solve two simultaneous equations to find the mass of a small pot and the mass of a large pot.

[3]

[N22/I/6]

12. Each day, Behnaz leaves home at the same time and cycles the same route to work.

On Monday, her average speed is 18 km/h and she arrives 4 minutes late.

On Tuesday, her average speed is 24 km/h and she arrives 6 minutes early.

- (a) The time taken to travel to work on Monday is represented by t minutes.

Write an expression, in terms of t , for the time, in minutes, taken to travel to work on Tuesday.

[1]

- (b) Form an equation in t to find the time taken to travel to work on Monday.

[3]

[N22/I/13]

13. Niluka thinks of an integer.

He multiplies it by 6 and subtracts 20.

He multiplies the answer by 2 and subtracts 30.

His final answer is greater than 1000.

- (a) Using x to represent Niluka's integer, write down an inequality in x .

[2]

- (b) Solve your inequality to find the smallest possible value of x .

[2]

[N22/I/14]

14. (a) The first four terms of a sequence are 8, 11, 14 and 17.

The sum of the first n terms of this sequence is given by $pn^2 + qn$.

- (i) When $n = 1$, $p + q = 8$.

Show that $4p + 2q = 19$.

[1]

- (ii) Solve $p + q = 8$

$$4p + 2q = 19.$$

[3]

- (b) The sum of the first n terms of a different sequence is given by $3n^2 + 7n$.

Find the 10th term of this sequence.

[2]

[N19/I/25]

15. One solution of the equation $kx^2 + (k + 1)x - 4 = 0$ is $x = -2$.

Find

- (a) the value of k ,

[1]

- (b) the second possible value of x .

[1]

[N18/I/3]

16. The points $(-2, 11)$ and $(4, -1)$ lie on the curve given by the equation $y = ax^2 + bx + 3$.

Use an algebraic method to find the values of a and b .

[4]

[N18/I/21]

1. (a) Solve $\frac{x}{7} + \frac{x-5}{3} = 1$. [2]
 (b) Solve the equation $x^2 + 9x - 16 = 0$ by completing the square.
 Give your solutions correct to two decimal places. [4]
 [N22/II/1(a), (c)]

2. (a) Solve the inequality $\frac{2x+5}{2} > 1 - 4x$. [2]
 (b) Solve the equation $\frac{3x}{x-4} + \frac{5}{2x+1} = 1$.
 Give your solutions correct to 2 decimal places. [5]
 [N21/II/1(a), (c)]

3. (a) Solve the inequality $\frac{2x+1}{2} \geq \frac{5-4x}{3}$. [2]
 (b) Solve these simultaneous equations.

$$6x - 3y = 16$$

$$9x + 2y = 11$$
 [3]
 [N20/II/1(a), (b)]

4. Solve $\frac{1}{x-3} + \frac{6}{x-1} = 2$. [4]
 [N19/II/1(d)]

5. Solve the equation $\frac{20}{x+1} = 2x + 5$. [3]
 [N18/II/1(c)]

6. (a) Solve the equation $\frac{2x-3}{4} + \frac{x}{3} = 3$. [2]
 (b) Solve these simultaneous equations.

$$4x - 3y = 18$$

$$6x + 2y = 1$$
 [3]
 [N16/II/1(b), (c)]

7. (a) Solve the equation $\frac{9}{x-4} = 2x - 1$. [3]
 (b) (i) Express $x^2 - 9x + 17$ in the form $(x+a)^2 + b$. [1]
 (ii) Hence solve the equation $x^2 - 9x + 17 = 0$, giving your answers correct to two decimal places. [3]
 [N15/II/1(c), (d)]

8. (a) These are the first four terms in a sequence.

11 17 23 29

(i) Find an expression, in terms of n , for the n th term of the sequence. [2]

(ii) Explain why it is not possible for a term in the sequence to be a multiple of 3. [1]

- (b) The n th term of a different sequence is given by $T_n = \frac{4n-1}{205-5n}$.

(i) Use the formula to find T_5 .

Give your answer as a fraction. [1]

(ii) The value of T_k can be simplified to $\frac{7}{32}$.

Find the value of k . [3]

(iii) Find the least value of n for which $T_n > 1$. [3]

[N18/II/4]

9. An empty water tank has a capacity of 3500 litres.

Pump A can fill the tank at a rate of x litres per minute.

Pump B can fill the tank at a rate of $(x - 10)$ litres per minute.

(a) Write down an expression, in terms of x , for the number of minutes it would take to fill the tank using pump A. [1]

(b) Write down an expression, in terms of x , for the number of minutes it would take to fill the tank using pump B. [1]

(c) It takes 21 minutes longer to fill the tank using pump B than it does using pump A.

Write down an equation to represent this information and show that it reduces to

$$3x^2 - 30x - 5000 = 0. \quad [3]$$

(d) Solve the equation $3x^2 - 30x - 5000 = 0$, giving your solutions correct to two decimal places. [4]

(e) Calculate how long it would take to fill the tank from empty using pump A and pump B together.

Give your answer in minutes and seconds, correct to the nearest ten seconds. [2]

[N18/II/6]

10. (a) Solve the inequality $\frac{x+4}{5} \geq \frac{2-x}{3}$. [2]

(b) Solve the equation $\frac{2}{x-2} + \frac{3}{x+4} = 1$. [3]

[N17/II/1(a), (e)]

11. An open box is in the shape of a cuboid.

The height of the box is 5 cm less than its length.

The width of the box is twice its height.

(a) Given that the length of the box is x cm, write down expressions, in terms of x , for the height and the width of the box. [2]

(b) The external surface area of the open box is 48 cm^2 .

Write down an equation to represent this information and show that it simplifies to

$$2x^2 - 15x + 13 = 0. \quad [3]$$

(c) Solve the equation $2x^2 - 15x + 13 = 0$. [3]

(d) Explain why one of the solutions in part (c) must be rejected as the length of the box. [1]

(e) Calculate the volume of the box. [2]

[N17/II/7]

12. A small business makes mugs with designs painted on them.

- (a) Some mugs are sold in a local shop, some in department stores and the rest are exported.

The mugs are divided between the local shop, department stores and export in the ratio 3 : 7 : 5.

One week, 140 more mugs are sent for export than to the local shop.

Work out the total number of mugs made in the week.

[2]

- (b) David and Maryam paint designs on the mugs.

- (i) David takes x seconds to paint one mug.

Write an expression, in terms of x , for the number of mugs he paints in an hour.

[1]

- (ii) Maryam takes 50 seconds less than David to paint one mug.

Write an expression, in terms of x , for the number of mugs she paints in one hour.

[1]

- (iii) One morning, David and Maryam each work for 4 hours.

Altogether they paint a total of 68 mugs.

Write down an equation in x to represent this information and show that it reduces to

$$17x^2 - 8050x + 180\,000 = 0.$$

[3]

- (iv) Solve the equation $17x^2 - 8050x + 180\,000 = 0$.

[3]

- (v) Find the number of mugs Maryam paints in one hour.

[2]

[N16/II/7]

13. Wei drives at an average speed of x km/h for 2 hours 40 minutes and then at an average speed of y km/h for 1 hour 20 minutes.

He drives a total distance of 240 km.

- (a) Write down an equation in x and y to represent this information and show that it simplifies to $2x + y = 180$. [1]

Hok drives at an average speed of x km/h for 1 hour 30 minutes and then at an average speed of y km/h for 2 hours 30 minutes.

He drives 10.5 km further than Wei.

- (b) Write down an equation in x and y to represent this information. [1]

- (c) Solve these two equations to find the value of x and the value of y . [3]

- (d) Calculate how much longer it would take Wei to drive 240 km at the slower speed than it would at the faster speed.

Give your answer in minutes and seconds, correct to the nearest second.

[2]

[N15/II/6]

